

INDEX

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DIV. _____

ROLL NO. _____

SCHOOL _____

Sr. No.	PAGE NO.	TITLE	DATE	TEACHER'S SIGN / REMARKS
1		Analyze and exploring the deep learning Platforms	24-07-25	He
2		Implement a classifier using open-source data set	7/8/25	He 11/11/25
3		Study of the classifiers with respect to Statistical parameters	7/8/25	He 11/11/25
4		Build a simple feed forward neural network to recognize hand written characters	14/8/25	He 11/11/25
5		Study of activation function and its role	28-8-25	He
6		Implement gradient descent & back propagation in deep neural network	09-9-25	He 11/11/25
7		Build a CNN model to classify Cats & dogs Images	16-7-25	He 11/11/25
8		Experiment using LSTM		He 11/11/25
9		Build a RNN		
10		Perform Compression on MNIST data set using Autoencoder		He 11/11/25
11		to implement experiment using Variational Autoencoder		
12		Implement DCGAN to generate Colour images	27/10	He 11/11/25
13		Understanding the architecture of Pre-trained model	27/10	
14		Pre-trained CNN model as feature extractor using TLM		
15		Implement a Yolo model to detect objects		He

Completed ~~He~~

13. understanding the architecture of pre-trained model

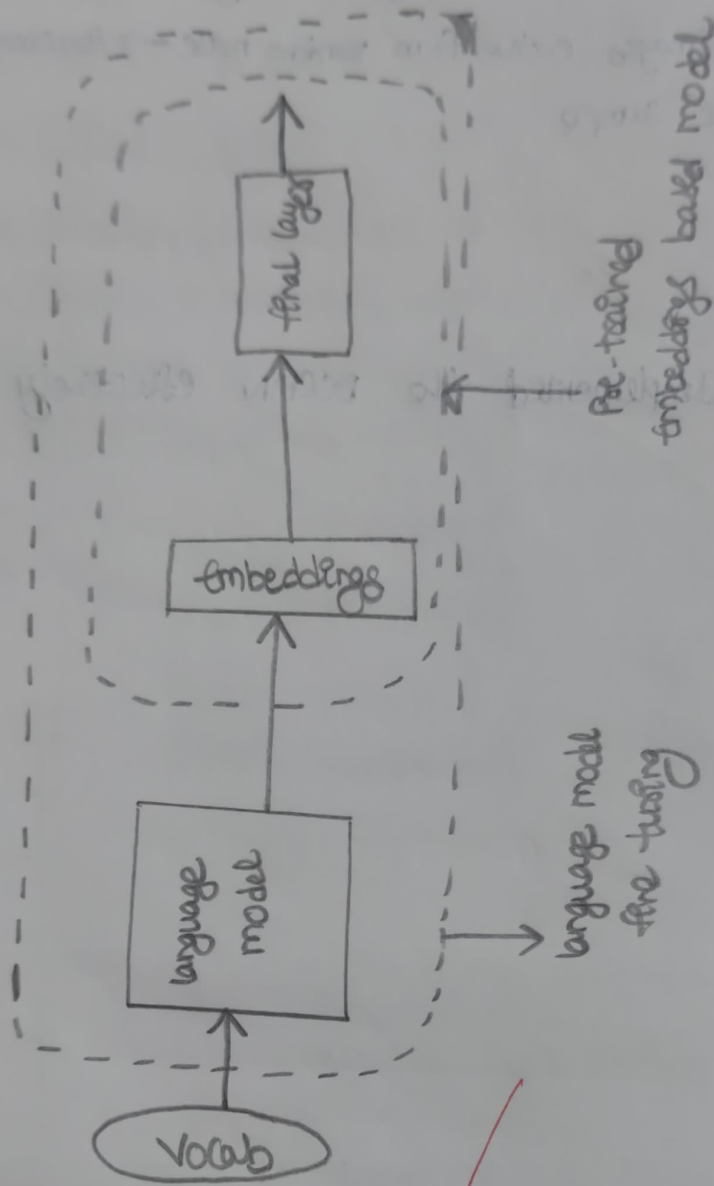
Aim :- To study and understand the architecture of pre-trained models. and analyze its layers, parameters and feature extraction capability.

Objective :-

- ① to load the pre-trained model from a library
- ② to understand layer structure, filter sizes and parameter count.
- ③ To visualize the architecture and identify Convolutional, Pooling and fully connected layers.
- ④ To understand how pre-defined (or) pre-trained models are used in transfer learning.

Pseudo Code :-

- ① Import Pytorch and torchvision libraries.
- ② Load a pre-trained model.
- ③ Print the model summary to view its architecture.
- ④ Iterate through each layer to observe
 - ① Convolutional layers
 - ② Pooling layers
 - ③ Batch normalization
 - ④ fully connected layers
- ⑤ Count and display total parameters





[1]

```
# LAB 13: Understanding the Architecture of a Pre-trained Model
# Dataset: CIFAR-10 (Open Source)
# Model: ResNet18 (Pre-trained on ImageNet)
# Algorithm
# Import VGG16
# Display summary
# Identify layers & parameters
####
from tensorflow.keras.applications import VGG16
model = VGG16()
model.summary()
```

Model: "vgg16"



Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,080
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,080
block3_pool (MaxPooling2D)	(None, 28, 28, 256)	0
block4_conv1 (Conv2D)	(None, 28, 28, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 14, 14, 512)	0
block5_conv1 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808



block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0
flatten (Flatten)	(None, 25088)	0
fc1 (Dense)	(None, 4096)	102,764,544
fc2 (Dense)	(None, 4096)	16,781,312
predictions (Dense)	(None, 1000)	4,097,000

Total params: 138,357,544 (527.79 MB)

Trainable params: 138,357,544 (527.79 MB)

Non-trainable params: 0 (0.00 B)

Output

total parameters : 138357544

trainable parameters : 138357544

layer names :

conv0 : conv0 2D

bn1 : batch normal 2D

relu : Relu

maxpool : maxpool 2D

layer1 : Sequential

layer2 : Sequential

layer3 : Sequential

layer4 : Sequential

avg pool : avg pool 2D

Observation

- * VGG16 : deep sequential network with 13 convolutional + 3 dense layers
- * Each pre-trained model system hierarchical image features efficiently

Result

Pre-trained models like VGG16, ResNet 50 and Inception V3 provide powerful feature.

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17